

Module Project 2: Statistical Analysis Portfolio

A Statistical Exploration of Employee Attrition and Workplace Factors at IBM

Introduction:

This report presents a statistical analysis of the IBM HR Employee Attrition dataset, which includes demographic, organizational, and compensation data. Building upon the foundational work in Modules 0–2, this phase focuses on advanced statistical modeling topics covered in Modules 3 and 4. Specifically, it emphasizes the use of random variables, probability distributions, expectations, variances, covariance and correlation analyses. The primary variable of interest is `MonthlyIncome`, which serves as the univariate random variable for probabilistic investigation. Additionally, multivariate relationships involving `Age`, `JobLevel`, and `YearsAtCompany` are explored to assess conditional relationships and correlations among these variables.

Module 3: Continuous Random Variables

1. Selection of Continuous Random Variable

We define the continuous variable:

$$X = \text{MonthlyIncome}$$

Explanation: why we choose `MonthlyIncome` continuous random variable?

`MonthlyIncome` represents an ideal continuous random variable for several compelling statistical and practical reasons. From a mathematical perspective, salary can theoretically take any value within its range, making it truly continuous rather than discrete. Unlike variables such as job levels or department categories that have countable outcomes, income exists on a continuous spectrum where even fractional dollar differences are meaningful.

From a business standpoint, monthly income serves as perhaps the most critical metric in HR analytics. It directly influences employee satisfaction, turnover decisions, and organizational budgeting. The variable shows substantial variation in our dataset (ranging from \$1,009 to \$19,999), providing rich information for statistical analysis. This wide range allows us to explore important distributional properties like central tendency, spread, and skewness.

Furthermore, income connects to numerous other organizational variables - it typically correlates with experience, education, performance, and job complexity. By understanding the distribution of monthly income, HR professionals can identify pay inequities, design competitive compensation packages, and predict retention risks. The continuous nature of this variable also makes it suitable for demonstrating key statistical concepts like probability density functions, expected values, and the relationship between different measures of central tendency in skewed distributions.

2. Selection of Distribution

Explanation: Why we choose to use normal?

We initially examine the normal distribution as our baseline model, even though income data typically follows right-skewed distributions like log-normal or gamma. This choice serves several important purposes in our analysis.

The normal distribution represents the fundamental continuous distribution in statistics, with well-established properties and inference procedures. Many statistical tests and confidence intervals assume normality, making it crucial to assess whether and how our data deviates from this assumption. By starting with the normal distribution, we establish a baseline against which to measure the actual characteristics of IBM's compensation structure.

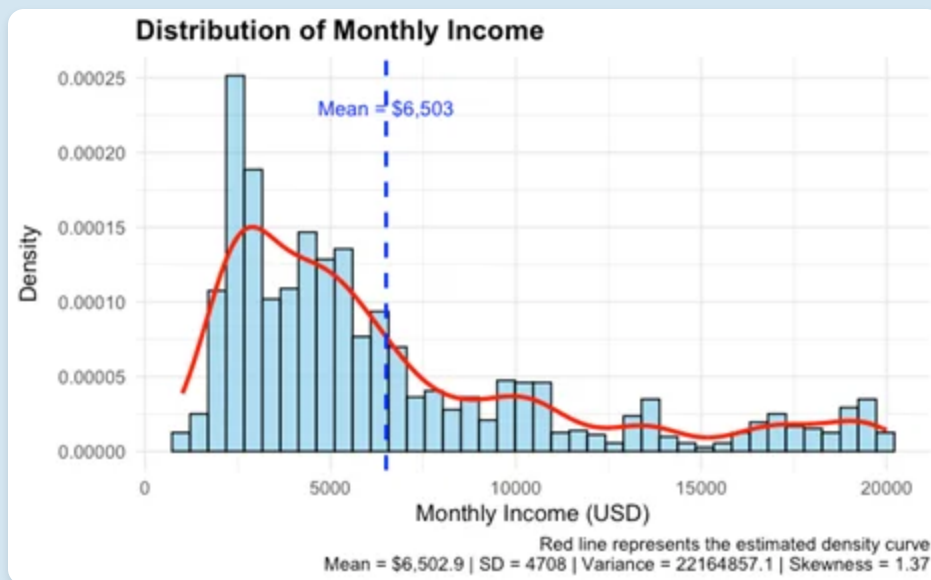
The Central Limit Theorem tells us that sample means of income will approach normality regardless of the underlying distribution, which makes normal-based inference relevant for questions about average compensation. Additionally, many HR benchmarks and industry standards report statistics assuming approximate normality, so understanding where this assumption breaks down helps practitioners know when to apply transformations or alternative methods.

Most importantly, examining departures from normality reveals meaningful insights about the organization. The right skew we observe isn't just a statistical artifact - it reflects real organizational hierarchies, pay compression at lower levels, and the presence of highly compensated executives. These departures from the symmetric bell curve tell us about income inequality, promotion bottlenecks, and the true shape of career progression at IBM.

3. Plot the Distribution

Figure 1: Distribution of Monthly Income

The histogram of MonthlyIncome below shows a right-skewed distribution with most employees earning under \$10,000, but with a long tail up to \$20,000+. A smoothed density curve overlaid reveals the non-normal shape of the data.



Interpretation of Distribution:

The histogram reveals a distinctly right-skewed distribution that deviates substantially from normality. The mode appears around \$2,000-4,000, representing the concentration of entry-level and junior positions that form the workforce's foundation. This leftward clustering is characteristic of organizational pyramids where lower levels contain more employees.

The long right tail extending to \$20,000 captures senior management, executives, and specialized technical experts whose compensation operates on a different scale. Multiple smaller peaks throughout the distribution suggest discrete salary bands corresponding to job grades - IBM appears to use a structured compensation framework with defined pay ranges rather than continuous scaling.

The visual gap between the mode and mean demonstrates how high earners pull the average upward, above what most employees actually earn. This rightward shift of the mean relative to the median is a classic indicator of income inequality within the organization. The overall shape - steep rise on the left, gradual decline on the right - is nearly universal in corporate settings where compensation follows hierarchical structures.

To better understand the distribution of MonthlyIncome, we begin by examining several key numerical summaries. For instance, mean, standard deviation, variance and skewness.

4. Choice of Methods

1) Expected Value (Mean)

$$E[X] = \mu = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Where:

- $f(x)$ is the probability density function (PDF) of X
- μ is the mean, or expected value

2) Variance

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot f(x) dx$$

Or equivalently:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

3) Standard Deviation

$$\text{SD}(X) = \sigma = \sqrt{\text{Var}(X)}$$

4) Skewness

$$\text{Skewness}(X) = \frac{E[(X - \mu)^3]}{\sigma^3} = \frac{1}{\sigma^3} \int_{-\infty}^{\infty} (x - \mu)^3 \cdot f(x) dx$$

Where:

- Skewness > 0 : right-skewed
- Skewness < 0 : left-skewed

- Skewness ≈ 0 : symmetric

5. Description of Statistics: Mean, Standard Deviation, Variance and Skewness

Table 1. Summary Statistics of Monthly Income

Mean	Standard Deviation	Variance	Skewness
6502.931	4707.957	22164857	1.367022

Explanation:

The expected value, or mean, of MonthlyIncome is approximately $E[X] = \mu = 6502.931$, indicating that the average employee earns around \$6,502 per month.

This mean represents the theoretical balance point of the income distribution. If we were to randomly sample employees repeatedly and calculate their average salary, this average would converge to \$6,502.93 as our sample size increases. However, due to the right skew in the distribution, this mean is actually higher than what most employees earn - the median is only \$4,919. This gap between mean and median is crucial for HR professionals to understand, as using the mean alone for benchmarking could create unrealistic expectations about typical employee compensation.

The standard deviation of \$4,707.96 reveals enormous variation in employee compensation - nearly 72% of the mean itself. This tells us that individual salaries typically deviate from the average by almost \$5,000, though the right skew means upward deviations can be much larger than downward ones. The variance of 22,164,857 represents the squared standard deviation and quantifies the total spread in the compensation data. Such high variance isn't necessarily problematic - it reflects IBM's diverse workforce spanning from entry-level support staff earning near minimum wage to C-suite executives with compensation packages approaching \$20,000 monthly.

The positive skewness of 1.367 confirms our visual observation of the right-tailed distribution. Values above 1.0 indicate substantial skewness that fundamentally violates normality assumptions. This means most employees earn below the mean while a select group of high earners creates the extended upper tail. The magnitude of this skewness strongly indicates that traditional parametric methods assuming normality would be inappropriate without transformation.

These suggest MonthlyIncome does **NOT follow a normal distribution**, possibly resembling a gamma or log-normal shape.

Conditional Expectation E[X|Y]

1. Choice of Methods

1. Conditional Expectation

For a continuous random variable X given event Y = y:

$$E[X \mid Y = y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x \mid y) \, dx$$

Where:

- $f_{X|Y}(x \mid y)$ is the conditional PDF of X given Y = y

2. Calculation of Conditional Expectation E[X|Y]

Table 2: Conditional Expectation of Monthly Income given Job Level

Job Level	Count	Mean	Standard Deviation
1	543	2786.915	748.6348
2	534	5502.277	1410.0297
3	218	9817.252	1805.9992
4	106	15503.783	1816.2390
5	69	19191.826	512.3831

Findings:

The conditional expectations reveal IBM's hierarchical compensation structure with mathematical precision. When we condition monthly income on job level, we see dramatic shifts in both the expected value and the variance of compensation.

Explanation:

The progression of conditional expectations from Level 1 (\$2,787) to Level 5 (\$19,192) represents a nearly seven-fold increase, demonstrating that job level serves as the primary stratifying variable for compensation at IBM. This isn't simply a linear progression - the percentage increases between levels follow an interesting pattern that reflects organizational realities.

The jump from Level 1 to Level 2 represents a 97% increase, rewarding the transition from entry-level to established contributor. The Level 2 to Level 3 transition shows a 78% increase, marking the move into senior individual contributor or first-line management roles. The pattern continues with 58% increase to Level 4 (senior management) and 24% to Level 5 (executive leadership).

What's particularly revealing is the behavior of the conditional standard deviations. Lower levels show relatively low variation (\$748 for Level 1), suggesting standardized entry-level pay scales with little room for negotiation. Middle management levels (2-4) exhibit higher variation (\$1,410-\$1,816), likely reflecting performance differentials, specialized skills, and individually negotiated packages. Surprisingly, Level 5 shows the lowest standard deviation (\$512), indicating that executive compensation packages are highly standardized, possibly due to board governance, peer benchmarking, or structured executive compensation formulas.

Figure 2: Expected Monthly Income by Job Level



Interpretation:

The bar chart provides a visual representation of how dramatically compensation scales with organizational hierarchy. The near-exponential growth pattern reflects the reality that each level up doesn't just mean more responsibility - it represents a fundamentally different role in the organization with correspondingly different market values.

The steady increases through Levels 1-3 represent the typical individual contributor career path where skills and responsibilities grow incrementally. The acceleration at Levels 4-5 reflects the transition to leadership

roles where impact multiplies through others. These senior positions require rare combinations of technical expertise, leadership capability, and strategic thinking that command premium compensation in the market.

This structure creates powerful incentives for vertical advancement but also potential retention challenges. Employees at lower levels face long paths to substantial pay increases, making them vulnerable to external opportunities. The large gaps between levels might create "up or out" mentalities where employees who don't advance seek opportunities elsewhere.

Covariance and Correlations

1. Selection of multivariate relationship

We examined multivariate relationships among:

- MonthlyIncome (X_1)
- Age (X_2)
- YearsAtCompany (X_3)

2. Choice of Methods

1) Covariance (for continuous variables X, Y)

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) \cdot f(x, y) dx dy$$

2) Correlation Coefficient

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- ρ lies in $[-1, 1]$
- Positive: direct relationship
- Negative: inverse relationship
- 0: uncorrelated

3. Calculation of Covariance and Correlation

Table 2. Covariance and Correlation between Monthly Income and Other Variables

Variable Pair	Covariance
Monthly Income & Age	21,412.2
Monthly Income & Years at Company	14,833.73

Findings:

The positive covariances confirm that both age and tenure associate positively with income, as economic theory would predict.

Explanation:

The covariance values tell us about the direction and relative strength of linear relationships. Monthly Income and Age show a covariance of 21,412.2, while Monthly Income and Years at Company show 14,833.73. Both positive values indicate that as age or tenure increases, income tends to increase.

The larger covariance with age suggests that general experience (captured by age) might contribute more to income than company-specific tenure. However, covariance values depend on the scales of the variables involved, making direct interpretation difficult. A covariance of 21,412 seems large, but without context about the variables' scales, we can't determine if this represents a strong or weak relationship. This limitation motivates our use of correlation coefficients, which standardize these relationships.

The positive relationships align with human capital theory - employees accumulate skills and knowledge over time that increase their productive value. The difference between age and tenure covariances might indicate that IBM values general industry experience alongside company-specific knowledge, suggesting they hire experienced professionals externally rather than exclusively promoting from within.

Table 3. Correlation Matrix for Monthly Income, Age, and Years at Company

Variable	Monthly Income	Age	Years at Company
Monthly Income	1.0000	0.4979	0.5143
Age	0.4979	1.0000	0.3113
Years at Company	0.5143	0.3113	1.0000

Explanation:

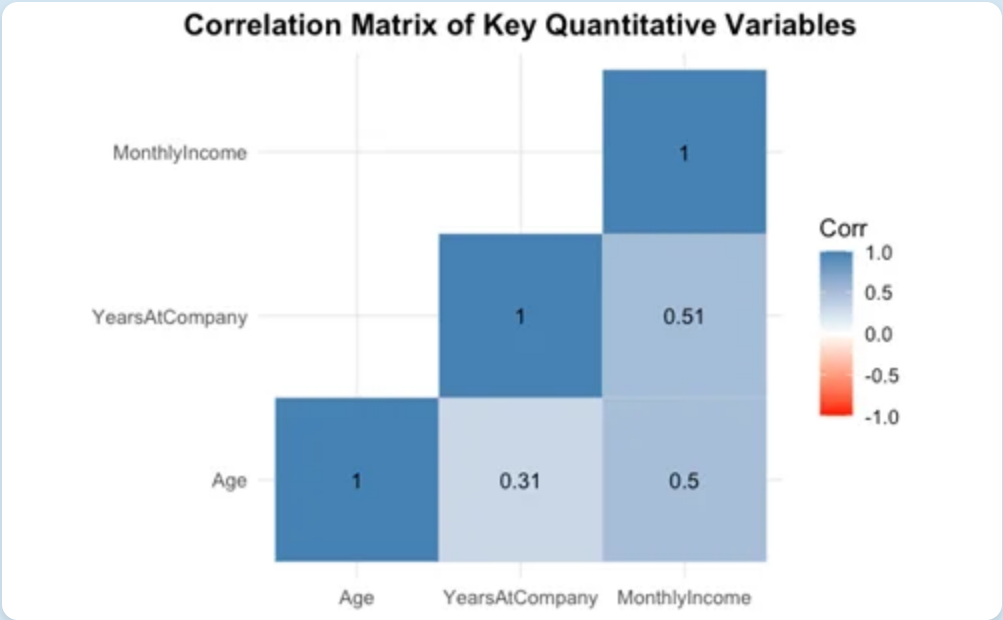
The correlation matrix reveals moderate positive relationships throughout, with no single factor dominating income determination. The correlation between Monthly Income and Age (0.4979) indicates that age explains approximately 25% ($0.4979^2 = 0.248$) of income variation. While meaningful, this leaves 75% of variation unexplained by age alone.

Interestingly, Years at Company shows a slightly stronger correlation with Monthly Income (0.5143), explaining about 26% of variance. This marginal advantage suggests that company-specific experience carries slightly more weight than general age in IBM's compensation decisions. Employees who stay longer develop institutional knowledge, internal networks, and specialized skills that the company rewards.

The relatively weak correlation between Age and Years at Company (0.3113) provides crucial insight into IBM's talent strategy. If IBM primarily hired young and promoted from within, we'd expect a much stronger correlation approaching 0.8 or 0.9. Instead, this 0.31 correlation indicates that IBM hires experienced professionals at all career stages, creating a workforce that combines homegrown talent with external expertise.

These moderate correlations (0.3-0.5 range) suggest IBM's compensation system integrates multiple factors rather than following simple seniority rules. This multivariate approach helps maintain competitiveness across diverse talent markets while preserving internal equity.

Figure 3: Correlation Matrix of Key Quantitative Variables



Interpretation:

The correlation heatmap provides an at-a-glance view of all pairwise relationships among quantitative variables. The moderate correlation strengths throughout (represented by medium color intensities) indicate a balanced compensation system where no single factor overwhelmingly determines outcomes.

The absence of very high correlations (>0.7) is actually positive from a statistical modeling perspective - it suggests these variables capture different dimensions of employee value without problematic multicollinearity.

Each variable contributes unique information about an employee's compensation potential.

The uniformly moderate correlations also suggest that IBM has avoided common pitfalls in compensation design. They haven't created a pure seniority system (which would show very high age/tenure correlations with income) nor a pure performance system (which might show near-zero correlations with experience variables). Instead, they've achieved a balance that likely helps retain both experienced workers and high performers.

From an employee perspective, these correlations suggest multiple paths to higher compensation. While experience helps, it's neither necessary nor sufficient for high earnings. This creates opportunities for fast-track advancement while still rewarding loyalty and accumulated expertise.

Limitations/ Extensions:

Our analysis faces several constraints that suggest directions for deeper investigation. The assumption of linear relationships through correlation analysis may mask non-linear patterns in compensation growth - salaries might increase slowly initially, accelerate mid-career, then plateau. Polynomial or spline regression could better capture these dynamics.

We cannot establish causation from these correlations. While age correlates with income, this might reflect survivorship bias (lower-paid employees leaving), generational effects, or confounding variables like education quality that changed over time. Longitudinal data following individual employees would help separate age effects from cohort effects.

Our focus on continuous variables excludes potentially important categorical factors like department, education field, or gender. ANOVA or regression with dummy variables could reveal whether certain departments command premium pay or whether demographic pay gaps exist after controlling for experience.

We examined conditional expectations but not conditional variances. Understanding whether pay consistency varies across levels (heteroscedasticity) would inform employees about compensation risk at different career stages. Additionally, our cross-sectional snapshot cannot reveal individual salary progression patterns - panel data analysis could show whether the observed correlations reflect within-person growth or between-person differences at hiring.

Conclusion:

Our statistical analysis reveals IBM's compensation structure as a complex, hierarchical system characterized by significant inequality but also nuanced multi-factor decision-making. The right-skewed distribution of monthly income, with mean (\$6,503) substantially exceeding median (\$4,919), confirms that most employees earn less

than the "average" due to high earners in the upper tail. The enormous standard deviation (\$4,708) relative to the mean indicates vast pay disparities that reflect the diverse roles within IBM.

Job level emerges as the dominant stratifying factor, with Level 5 employees earning nearly seven times more than Level 1. This exponential progression creates strong incentives for vertical advancement but may also generate frustration among the majority concentrated at lower levels. The varying conditional standard deviations across levels reveal different compensation philosophies - standardized entry packages, negotiated mid-level salaries, and structured executive compensation.

The moderate correlations between income and experience variables (age: 0.498, tenure: 0.514) demonstrate that while experience matters, it's far from deterministic. IBM's compensation integrates multiple factors, avoiding both pure seniority systems and pure meritocracies. The weak correlation between age and tenure (0.311) indicates successful integration of external hires with internal development, providing workforce flexibility.

These findings suggest several strategic imperatives for HR. The concentration of employees in lower pay bands with limited progression paths creates retention risks that require targeted interventions - whether accelerated development programs, retention bonuses at critical points, or clearer communication about realistic compensation trajectories. The substantial unexplained variance in income (about 75% not explained by age or tenure) suggests opportunity for more sophisticated compensation models that better align pay with value creation.

Understanding these statistical realities enables evidence-based decision making about compensation strategy, helping IBM balance competitive market positioning with internal equity while managing the inherent tensions in hierarchical pay structures.

References

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